FLEET & SHORE PERSONNEL TRAINING

Electrical Safety and Practices Onboard Training

Recognising electrical hazards onboard, working safely under a permit-to-work and lockout-tagout regime, and responding correctly to an electrical incident or shock.

SOLAS Ch. II-1

IEC 60092

STCW

Company SMS

COURSE MAP

What We Will Cover

01 **Electrical Hazards Onboard**
Shock, arc flash, fire and explosion risk in the marine environment

02 **Roles, Rules & Responsibilities**
Who may work on electrical systems and under what authority

03 **Safe Isolation & Lockout-Tagout**
The permit-to-work sequence that makes a circuit safe to touch

04 **PPE & Safe Tools for Electrical Work**
Insulating protection and correctly rated equipment

05 **High-Voltage System Precautions**
Extra controls for systems above 1000 V

06 **Portable Equipment & Insulation Testing**
Inspecting leads, tools and megger testing

07 **Emergency Response to Electrical Incidents**
Rescuing a shock casualty and reporting the event

SECTION 1

Electrical Hazards Onboard

Why a ship's electrical installation is a uniquely demanding environment, and what can go wrong when it is not respected.

01

WHY IT MATTERS

A Uniquely Demanding Environment

Salt air, moisture, vibration and confined compartments make marine electrical work more hazardous than an equivalent job ashore.

- Ships combine high-power generation and distribution with damp, salt-laden air, vibration and constant motion — all of which degrade insulation over time.
- Metal hulls, decks and structures are excellent conductors, so a person in contact with them has almost no protective resistance to earth.
- Confined spaces such as switchboard rooms, steering gear flats and cargo holds concentrate risk: escape routes are limited and rescue is harder.
- Electrical energy is invisible and silent — there is no warning sign before contact, unlike many other onboard hazards.
- Most electrical accidents onboard trace back to a shortcut: working live to save time, bypassing an interlock, or skipping isolation.

No Second Chances

Electricity does not injure gradually — a fault, a slip of a tool or a missed isolation step can be fatal in an instant. Every precaution in this course exists because someone, somewhere, did not take it.

TYPES OF HAZARD

Four Ways Electricity Causes Harm

Each hazard type calls for a different set of controls, covered later in this course.

Electric Shock

Current passing through the body can disrupt the heart's rhythm, cause burns, or cause a fall from a ladder or platform.

Arc Flash / Arc Blast

A short circuit can release an intense flash of heat and pressure, causing severe burns and hearing damage in a fraction of a second.

Electrical Fire

Overloaded cables, poor connections or damaged insulation are a leading cause of fires in machinery spaces and accommodation.

Explosion in Hazardous Areas

A spark from non-certified equipment in a gas-hazardous zone (tanker decks, paint stores) can ignite a flammable atmosphere.

THE PHYSIOLOGICAL EFFECT

How Little Current It Takes

The effect of AC current on the human body escalates sharply over a narrow range of milliamps (mA).

1 mA

Threshold of perception — a slight tingling sensation, usually harmless on its own

10-15 mA

Muscular contraction begins; a person can lose the ability to let go of the conductor

50-100 mA

Risk of ventricular fibrillation and cardiac arrest with exposure of only a few seconds

SECTION 2

Roles, Rules & Responsibilities

02

Electrical work onboard is authorised, not open to anyone with a screwdriver — who may do what, and under whose permission.

AUTHORISATION

Who May Work on Electrical Systems

Company SMS procedures, backed by STCW competence requirements, define who is authorised to carry out electrical work.

- Only the Chief Engineer, Electro-Technical Officer (ETO) or a person they specifically authorise may carry out work on live or de-energised electrical equipment.
- STCW Chapter III sets the minimum standard of competence for officers responsible for electrical, electronic and control engineering.
- The Master retains overall responsibility for the ship's safety and must be informed of any work affecting essential electrical services.
- Shore electricians or contractors working aboard must be briefed on the ship's isolation procedures and supervised by ship's staff.
- No crew member should attempt electrical work beyond their training or authorisation, even for what looks like a simple repair.

When in Doubt, Stop

If a task is not clearly within your authorisation and competence, stop and refer it to the ETO or Chief Engineer — assumptions about 'simple' electrical jobs are how accidents happen.

REGULATORY BASIS

The Rules Behind This Course

Electrical safety onboard sits within a wider framework of statutory and company requirements.

Instrument	What It Covers
SOLAS Chapter II-1, Part D	Minimum requirements for electrical installations on ships
IEC 60092 series	International standard for electrical installations in ships
STCW Code, Chapter III	Competence standards for engineer officers and ETOs
Company SMS / Permit-to-Work Procedure	Ship-specific isolation, permit and reporting requirements

SECTION 3

Safe Isolation & Lockout-Tagout

03

The disciplined sequence that turns a live, dangerous circuit into one that is verified safe to work on.

THE SEQUENCE

Six Steps to Safe Isolation

Skipping or reordering these steps is the single most common cause of electrical accidents during maintenance.

- Identify the correct circuit and equipment from the ship's electrical drawings — never assume from memory or from a label alone.
- Switch off and isolate the circuit at the source, opening the circuit breaker or disconnector, not just a local switch.
- Lock the isolation point with a personal padlock and attach a signed, dated warning tag stating who is working and why.
- Prove the circuit dead using a voltage tester rated for the system, and prove the tester itself works before and after on a known live source.
- Discharge any stored energy — capacitors, charged cables — and apply earthing connections where the system requires them.
- Only then may work begin; on completion, remove tools, confirm the area is clear, and remove the lock/tag only after a final check.

Test, Don't Trust

A breaker position, a label or another person's word is not proof a circuit is dead. Always prove it dead yourself with a tested instrument, immediately before touching any conductor.

THE TOOLKIT

Lockout-Tagout Equipment

A ship should carry a dedicated, well-stocked LOTO kit, checked periodically as part of the SMS.

Personal Padlocks

Individually keyed to each worker; only that worker holds the key, and only they may remove their own lock.

Warning Tags

State the worker's name, date, time and reason for isolation; attached alongside the lock at every isolation point.

Multi-Lock Hasps

Allow several workers on the same job to each apply their own lock to a single isolation point.

Permit-to-Work Form

Documents the job scope, isolation points, hazards and required PPE, signed off by the authorising officer.

DISCIPLINE

Lockout-Tagout in Practice

The value of LOTO depends entirely on it being followed exactly, every time.

Do

- Isolate at every possible energy source, including back-feeds and standby supplies
- Keep your padlock key on your person at all times while the job is open
- Re-verify the circuit is dead if you leave and return to the job
- Close out the permit and remove your lock only after work is fully complete

Don't

- Don't ever remove or cut another worker's lock, for any reason
- Don't rely on a switch position alone as proof of isolation
- Don't leave a job partially isolated and unattended
- Don't bypass an interlock or safety cover to 'just take a quick look'

SECTION 4

PPE & Safe Tools for Electrical Work

04

The insulating barrier between a worker and a fault — equipment that must be correctly rated, inspected and used.

PERSONAL PROTECTIVE EQUIPMENT

Rated Protection for Electrical Work

Standard workwear does not protect against electrical hazards — dedicated, rated PPE is required.

Insulating Gloves

Rubber gloves rated to the working voltage, inspected before every use and periodically pressure-tested.

Insulating Mats

Placed at switchboards and panels to insulate the worker from the deck, a conductive surface.

Arc-Rated Clothing

Flame- and arc-resistant coveralls and face shields for work near live switchboards or high-fault-current equipment.

Insulated Hand Tools

Screwdrivers and pliers with certified insulated handles, kept exclusively for electrical work.

INSPECTION & USE

Before You Rely On It

PPE only protects if it is in good condition and matched to the job.

- Visually inspect gloves and mats for cuts, punctures or perishing before every use — a damaged item must be withdrawn from service immediately.
- Confirm the voltage rating of gloves and tools matches or exceeds the system being worked on; never assume one rating fits all jobs.
- Remove rings, watches and other conductive jewellery before any electrical work, live or isolated.
- Store insulating PPE away from sunlight, oil and sharp edges, in its dedicated bag or case between uses.
- A voltage-rated tester and proving unit are as much a part of the PPE set as gloves — never work without a means to prove dead.

PPE Is the Last Line

PPE protects if isolation fails or a fault occurs — it is never a substitute for proper isolation. Treat every conductor as live until you have proved otherwise.

SECTION 5

05

High-Voltage System Precautions

Ships with HV generation and distribution — above 1000 V — carry additional risk and additional controls.

WHY HV IS DIFFERENT

Extra Controls for Extra Risk

Larger vessels, particularly those with electric propulsion, carry systems where an arc flash can be catastrophic.

- HV switchgear is normally operated only by specifically authorised and trained personnel, never by general watchkeeping staff.
- A documented safe-access and switching procedure governs opening HV compartments, including a minimum time delay after isolation for capacitors to discharge.
- Approach distances to exposed HV conductors are strictly defined; only qualified persons with correct PPE may cross them.
- HV systems require earthing devices to be applied and visually confirmed before any compartment is entered for maintenance.
- Arc-flash risk assessments and warning labels at HV panels state the required PPE category and safe working distance for that specific panel.

Two-Person Rule

HV switching and access should never be carried out by one person alone — a second, trained person must be present to assist and to raise the alarm if something goes wrong.

COMPARING THE RISK

Low Voltage vs High Voltage Onboard

Both require isolation and PPE, but HV systems demand a materially stricter regime.

Aspect	Low Voltage (< 1000 V)	High Voltage (> 1000 V)
Who may operate	Authorised engine department staff	Specifically HV-trained and certified personnel only
Arc flash energy	Generally limited	Can be severe; arc-rated PPE mandatory
Approach distance	Contact-based hazard	Danger exists even without direct contact
Isolation proving	Voltage tester	Voltage tester plus mandatory earthing device

KEY REFERENCE POINTS

Numbers Every HV-Authorised Person Should Know

These figures anchor the extra precautions HV systems demand.

1000 V

The threshold above which a system is classed as high voltage under IEC 60092

2

Minimum number of trained persons present for any HV switching operation

0

Tasks that justify bypassing an HV interlock or working without proven isolation

SECTION 6

06

Portable Equipment & Insulation Testing

The everyday tools, leads and machinery that cause the most frequent — if less dramatic — electrical incidents.

EVERYDAY RISK

Portable Tools, Leads and Extension Cables

Portable electrical equipment sees rough handling and harsh conditions, making regular inspection essential.

- Visually inspect plugs, cables and casings before every use for cracks, exposed conductors or damaged earth pins.
- Route extension leads clear of walkways, hot surfaces and pinch points; never run them through watertight doors or hatches when closed.
- Use only equipment appropriate to the space — flameproof or intrinsically safe tools in hazardous (gas) areas.
- A residual current device (RCD/ELCB) should protect portable equipment circuits, particularly in damp spaces such as tank tops and deck areas.
- Damaged portable equipment must be tagged out of service and removed for repair or disposal — never used 'just this once'.

Reading a Megger Test

Periodic insulation resistance (IR) testing catches deteriorating insulation before it becomes a fault.

1 MΩ

Widely used minimum acceptable insulation resistance for low-voltage shipboard circuits

500 V

Typical test voltage applied by a megger for low-voltage circuit testing

↓ Trend

A falling IR reading over successive tests warns of degrading insulation, even above the minimum

TEST & MAINTENANCE EQUIPMENT

Keeping the Fleet's Wiring Trustworthy

Regular testing, logged in the PMS, is what turns 'it worked last time' into a verified fact.

Insulation Resistance Tester (Megger)

Applies a test voltage to measure insulation resistance to earth on isolated circuits and motor windings.

Voltage Proving Unit

Confirms a circuit is dead before work begins and live again after work is complete.

Earth Continuity Tester

Verifies the earth path of equipment casings and portable tools is intact and low-resistance.

PMS Test Records

Every insulation and continuity test result is logged against the equipment in the planned maintenance system.

SECTION 7

Emergency Response to Electrical Incidents

07

What to do — and not do — in the first seconds after witnessing an electric shock or an electrical fire.

FIRST RESPONSE

Finding a Person in Contact With a Live Conductor

The rescuer's own safety comes first — a second casualty helps no one.

Do

- Isolate the supply immediately if the switch or breaker is safely accessible
- If isolation is not possible, use a dry, non-conductive object to separate the casualty from the source
- Call for help and raise the alarm before or while attempting rescue
- Once safe, check breathing and begin CPR immediately if the casualty is not breathing normally

Don't

- Don't touch the casualty directly while they remain in contact with a live conductor
- Don't use a metal or wet object to try to free the casualty
- Don't assume a low-voltage source is 'safe enough' to touch directly
- Don't move a casualty with a suspected spinal injury unless remaining in place is more dangerous

AFTER THE RESCUE

Casualty Care and Reporting

Immediate care and accurate reporting both influence the outcome, for the casualty and for preventing a recurrence.

- Every electric shock casualty should be assessed by the ship's medical officer or via Telemedical Assistance Service, even if they appear to recover quickly — burns and cardiac effects are not always immediately visible.
- Treat any visible burns as per the ship's medical guide; electrical burns are often deeper than they first appear on the skin.
- For an electrical fire, isolate the supply before applying any extinguishing agent, and use only a CO_2 or dry-powder extinguisher rated for electrical fires — never water on live equipment.
- Preserve the scene and the failed equipment where practicable, to support the Company's investigation into root cause.
- Report the incident through the ship's SMS without delay, and notify the Company DPA for any incident involving injury or a near miss.

Every Near Miss Counts

A shock that caused no visible injury is still a reportable near miss — investigating it can prevent the next incident from being far more serious.

REPORTING CHAIN

Who Needs to Know, and How Fast

A clear, fast reporting chain ensures the right support reaches the casualty and lessons reach the wider fleet.

Step	Action	Responsible
Immediate	Isolate supply, rescue casualty, raise the alarm	First person on scene
Within minutes	First aid / CPR, request medical guidance	Duty officer / medical officer
Within hours	Notify Master and Company DPA; secure the scene	Master / Chief Engineer
Post-incident	Submit SMS report; root-cause investigation	Master with Company support

COURSE COMPLETE

Every isolation you prove, and every shortcut you refuse, is a shock or a fire that never happens.

Company DPA — 24/7 emergency line

Chief Engineer / ETO — onboard electrical authority

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